

Zero probabilities and Laplace smoothing

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Example: predicting whether to play or not

	Outlook		Temperature		Humidity		Windy		Play	
	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
Sunny	2	3	Hot	2	2	High	3	4	6	2
Overcast	4	0	Mild	4	2	Normal	6	1	3	3
Rainy	3	2	Cool	3	1					
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5		
Rainy	3/9	2/5	Cool	3/9	1/5					

Test instance $\tilde{X} = (\text{Outlook} = \text{sunny}, \text{Temp} = \text{cool}, \text{Humidity} = \text{high}, \text{Windy} = \text{true})$

$$P(\text{Play=no} \mid \tilde{X}) \propto P(\tilde{X} \mid \text{Play=no})P(\text{Play=no})$$

Probability estimations

$$P(x_i = a \mid C = c) = \frac{n(a, c)}{N(c)},$$

where

$n(a, c)$ is the number of training objects in class c with $x_i = a$,

$N(c)$ is the total number of training objects in class c

Example: predicting whether to play or not (zero probabilities)

	Outlook		Temperature		Humidity		Windy		Play	
	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
Sunny	2	3	Hot	2	2	High	3	4	9	5
Overcast	4	0	Mild	4	2	Normal	6	1		
Rainy	3	2	Cool	3	1					
Sunny	2/9	3/5	Hot	2/9	2/5	High	3/9	4/5	9/14	5/14
Overcast	4/9	0/5	Mild	4/9	2/5	Normal	6/9	1/5		
Rainy	3/9	2/5	Cool	3/9	1/5					

Test instance $\bar{X} = (\text{Outlook}=\text{overcast}, \text{Temp}=\text{cool}, \text{Humidity}=\text{high}, \text{Windy}=\text{true})$

$$P(\text{Play}=\text{no} \mid \bar{X}) \propto P(\bar{X} \mid \text{Play}=\text{no}) P(\text{Play}=\text{no})$$

$$= P(\text{Outlook}=\text{overcast} \mid \text{Play}=\text{no}) \times P(\text{Temp}=\text{cool} \mid \text{Play}=\text{no})$$

$$\times P(\text{Humidity}=\text{high} \mid \text{Play}=\text{no}) \times P(\text{Windy}=\text{true} \mid \text{Play}=\text{no}) \times P(\text{Play}=\text{no})$$

$$= 0 \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0$$

Zero probabilities

- ▶ **Issue:** If the feature value a_i does not co-occur with a class value c , then the corresponding probability estimation will be 0.
- ▶ For example: Given Outlook = overcast, the probability of Play = no is 0/5. The other features will be ignored as the final result will be multiplied by 0.
- ▶ This is bad for our 4 feature dataset, but terrible for (say) a 1000 feature dataset.
- ▶ In text classification, we often encounter situations where a feature does not occur in a particular class.

Laplace smoothing

- ▶ We can “borrow” some probabilities from high probability features and distribute them among zero probability features to avoid having features with zero probabilities.
- ▶ This is called *smoothing*.
- ▶ There are numerous smoothing techniques based on different policies. As long as the total probability mass remains unchanged any policy of probability reassignment is valid.
- ▶ A popular method is called **Laplace smoothing**. For feature x_i and class c the probability estimates will be updated as follows:

$$P(x_i = a \mid C = c) = \frac{n(a, c) + 1}{N(c) + m_i} \quad \text{for every value } a \text{ of feature } x_i$$

where $n(a, c)$ is the number of training objects in class c with $x_i = a$, $N(c)$ is the total number of training objects in class c , and m_i is the number of possible values of feature x_i .

Laplace smoothing: before

$$P(x_i = a | C = c) = \frac{n(a, c) + 1}{N(c) + m_i} \quad \text{for every value } a \text{ of feature } x_i$$

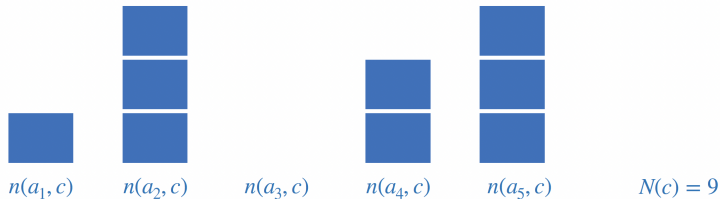


Figure: Laplace smoothing: before

Laplace smoothing: after

$$P(x_i = a \mid C = c) = \frac{n(a, c) + 1}{N(c) + m_i} \quad \text{for every value } a \text{ of feature } x_i$$

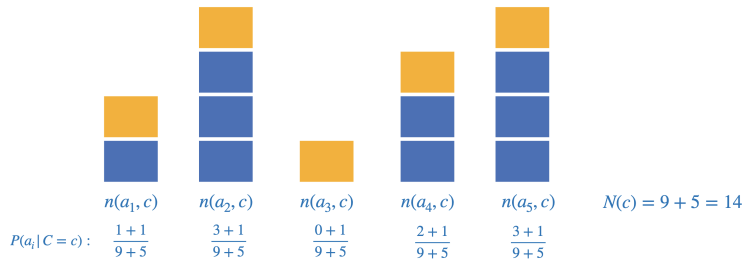


Figure: Laplace smoothing: after